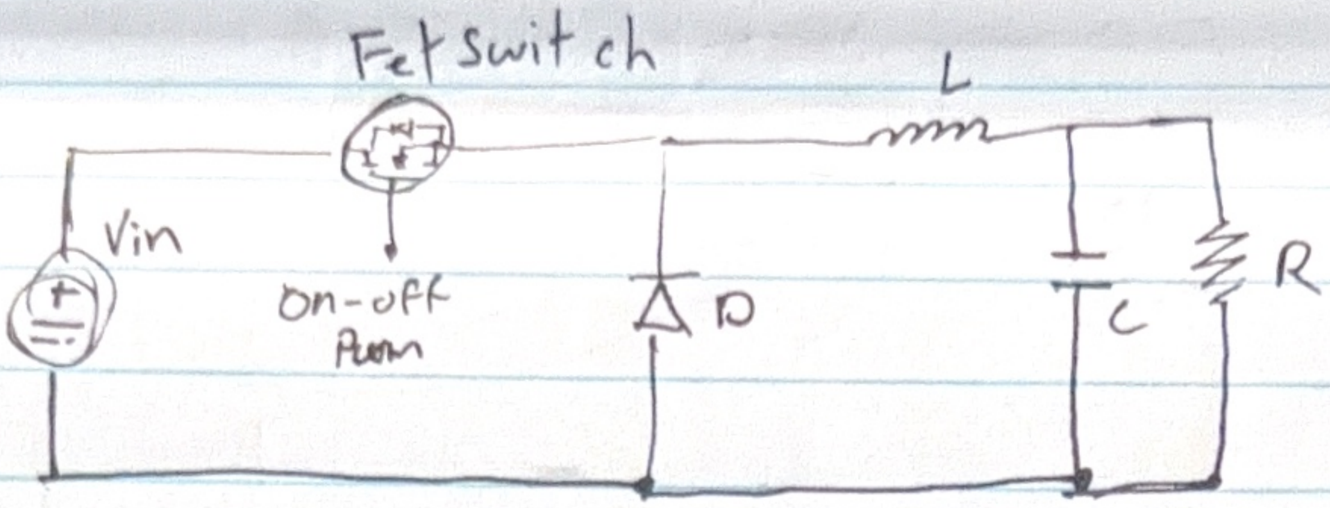
Buck converters DC/DC



More efficient

Ideal

$$V_{out} = D \cdot V_{in}$$

With Loss Elements

$$V_{out} = \frac{D \cdot V_{in} - (1-D)V_0}{1 + D \cdot \frac{R_{on}}{R} + \frac{R_L}{R}}$$

Discontinuous

$$V_{out} = V_{in} \cdot \frac{2}{1 + \sqrt{1 + \frac{4 \cdot k}{D^2}}}, \quad K = \frac{2 \cdot L}{R \cdot T}$$

Limitations

Unidirectionality

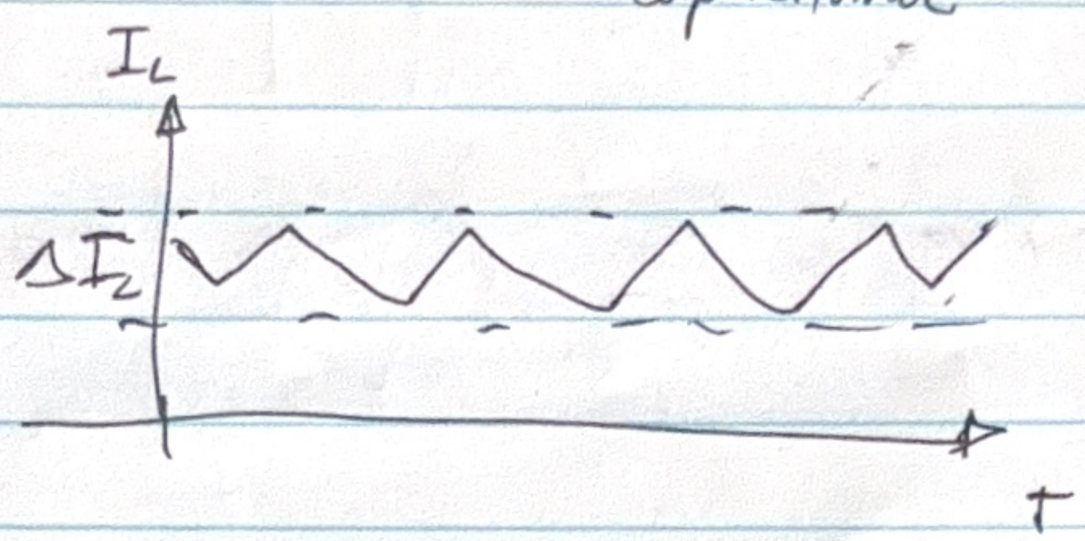
Increasing switching frequency is one way of shrinking inductors and capacitors

- current and voltage ripples are inversly proportional to frequency

Voltage Ripples: $\Delta V_{out} = \frac{\Delta I_L}{8 \cdot C \cdot f_{sw}}$

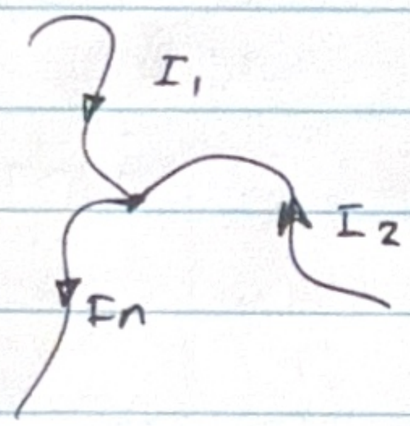
Interleaving

Capacitance



High input - output ratios
- tend to enter DCM unless L is a high value

KCL



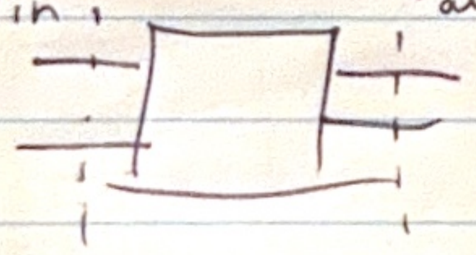
$\sum_j I_j = 0$

Average → $\langle \sum_j I_j \rangle = 0$

$\sum_j \langle I_j \rangle = 0$

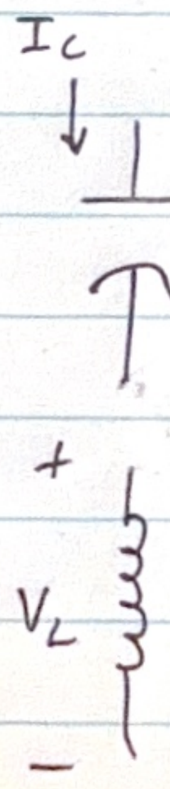
if lossless

P_{in} P_{out} $P_{in} = P_{out}$



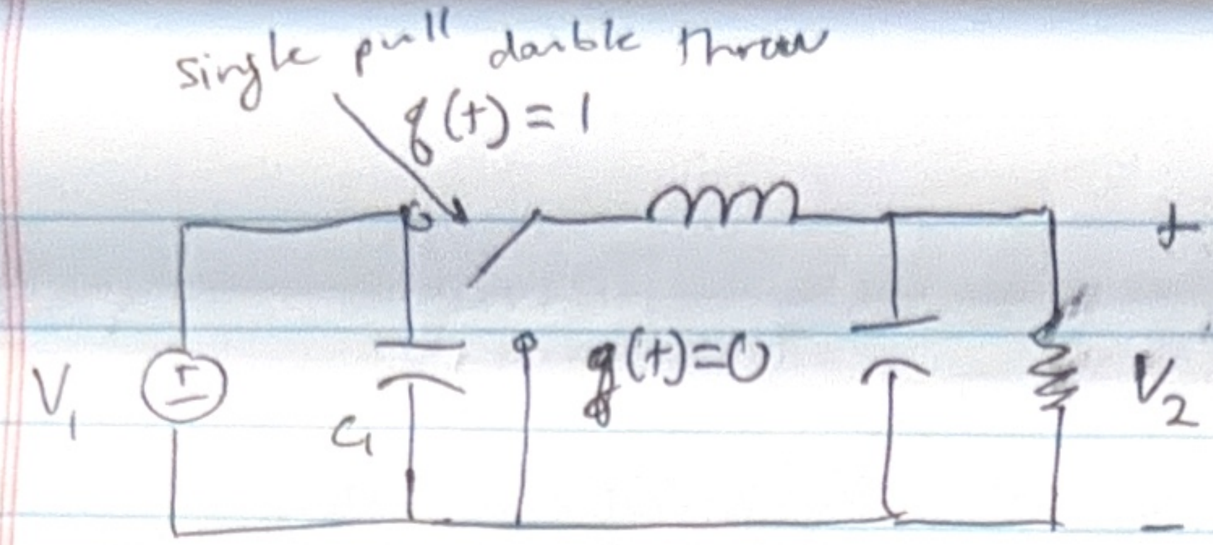
KVL $\sum_j \langle V_j \rangle = 0$

In PSS

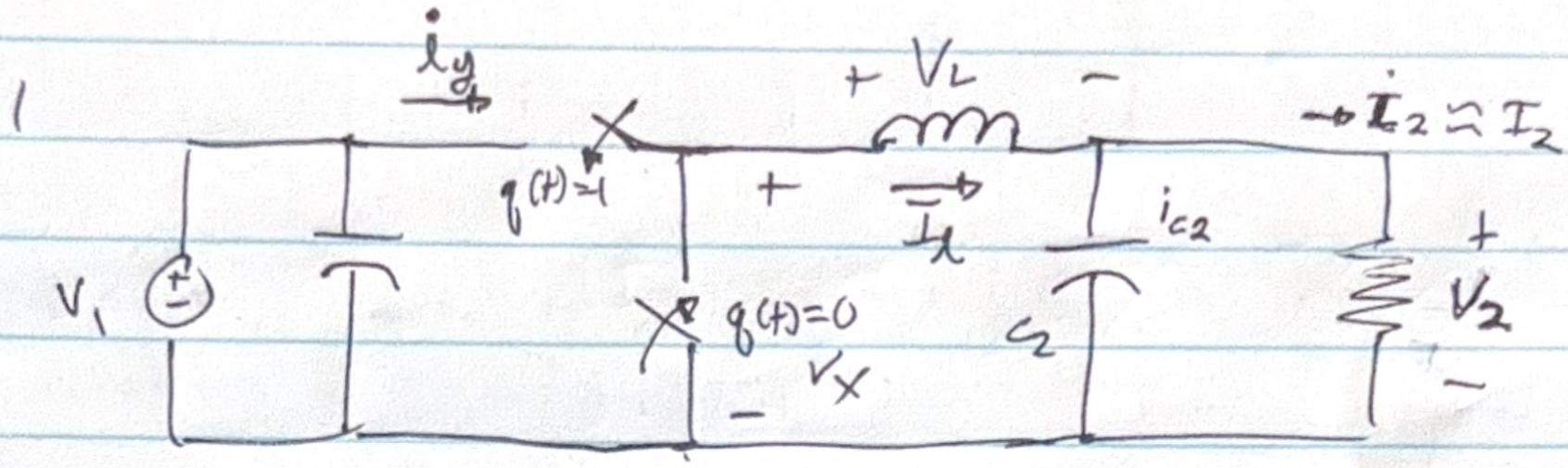


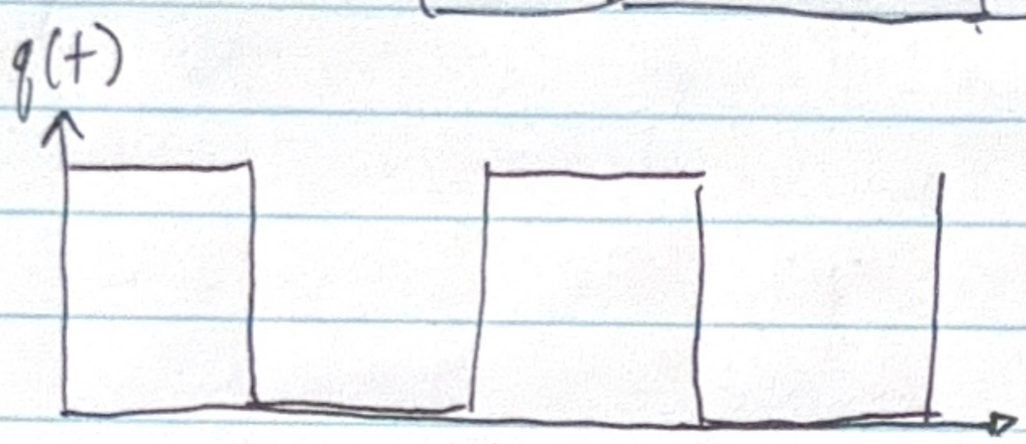
$\langle I_C \rangle = 0$

$\langle V_L \rangle = 0$



Practical

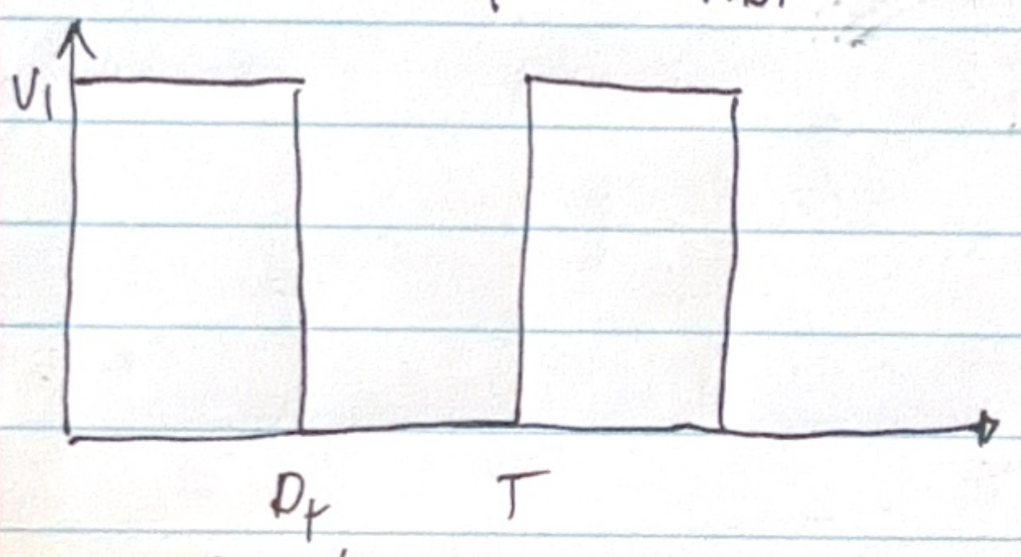




lossless

$$\boxed{V_2 = D \cdot V_1}$$

$$\boxed{I_y = D \cdot I_2} = I_1$$

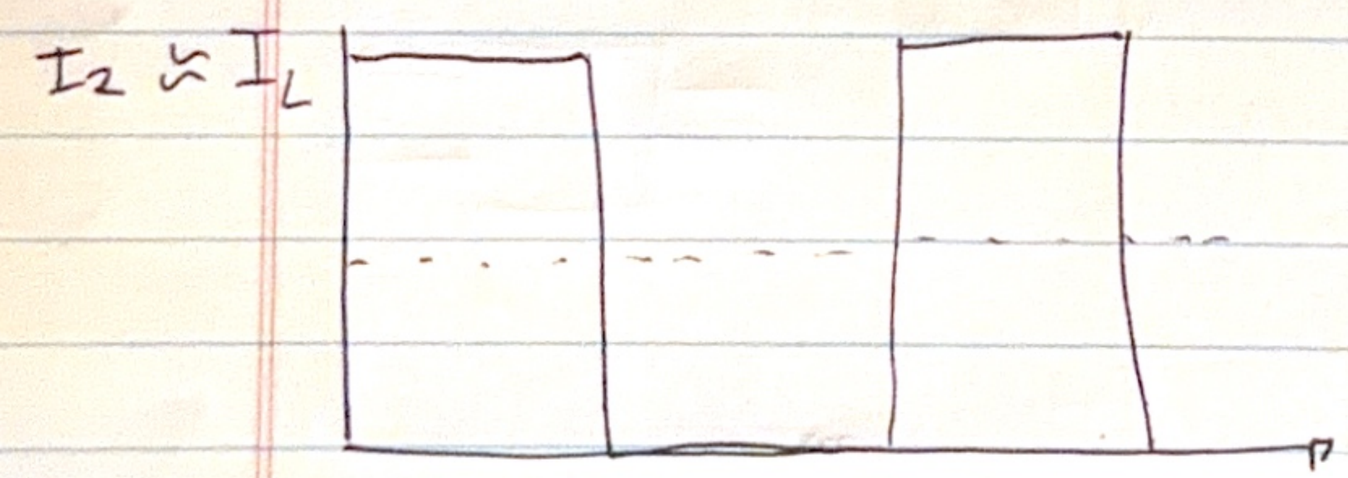


$$V_x - V_L - V_2 = 0$$

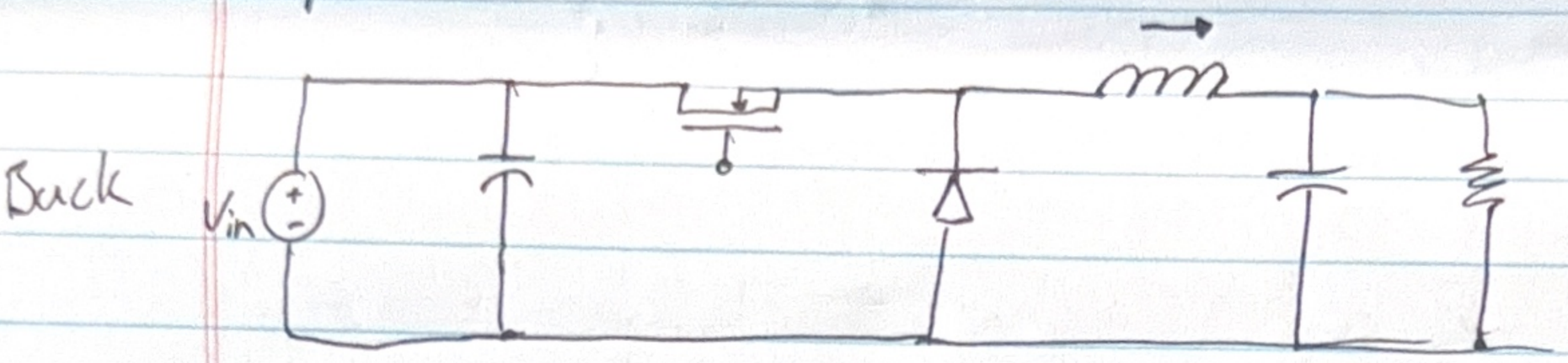
$$\langle V_x \rangle - \langle V_L \rangle - \langle V_2 \rangle = 0$$

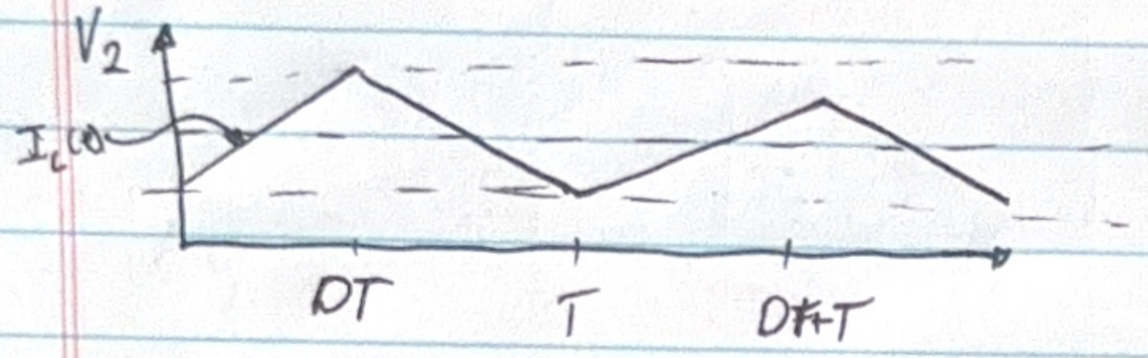
$\langle V_L \rangle$ ↑ 0 in Pss

$$\langle V_2 \rangle = \langle V_x \rangle$$



Implementation:

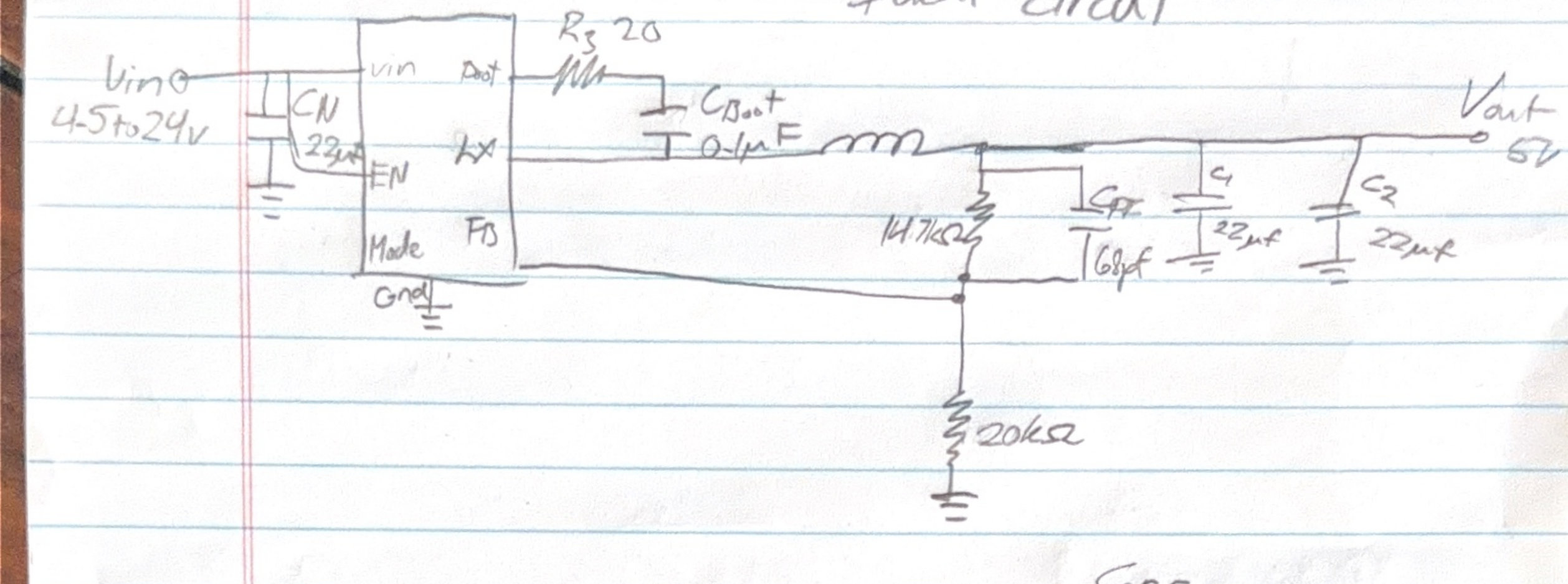




$\langle V_2 \rangle = \frac{1}{2} D \cdot V_x$

RT6255B Buck ic 5V out

Ideal circuit



C_{FF} 22 to 68

Inductor selection: $L = \frac{V_{out} \times (V_{in} - V_{out})}{V_{in} \times f_{sw} \times \Delta I_L}$

$$\Delta I_L = \frac{V_{out} \times (V_{in} - V_{out})}{V_{in} \times f_{sw} \cdot \Delta I_L}$$

Chose based on valley

$I_{L(valley)} = I_{out(max)} - \frac{1}{2}\Delta I_L$

Inductor saturation current must be over 4.5A

$$L = \frac{5 \times (12-5)}{12 \cdot 500 \times 10^5 \cdot 1.5} = \boxed{4.7\mu H}$$

Cap DC bias.

- must size up from 0603 on the 22μF caps or it will loose around 80% of its capacitance due to bias

Non Ideal circuit with 4 1210 22μF cap

